

Batu Ozturkler

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RESEARCH AND PRODUCT INTERESTS

I am interested in AI research with meaningful product impact. My recent work focuses on **confidence estimation and hallucination detection in LLMs, LLM post-training and fine-tuning, and agentic workflows with LLMs**. During my PhD, I worked on diffusion models, probabilistic reasoning with LLMs, computer vision, robustness under distribution shifts, memory-efficient learning, self-supervised learning, and medical/computational imaging.

EDUCATION

Stanford University <i>Ph.D. candidate in Electrical Engineering; GPA: 3.99/4.00</i> - Advisor: Prof. John Pauly and Prof. Mert Pilanci - Thesis: Efficient and robust deep learning for medical imaging and natural language processing	Sep 2019 – Jan 2024 Stanford, CA
Stanford University <i>Master of Science in Electrical Engineering; GPA: 3.99/4.00</i>	Sep 2019 – June 2021 Stanford, CA
Middle East Technical University (METU) <i>Bachelor of Science in Electrical Engineering; GPA: 3.96/4.00, Rank: 3/413</i>	Aug 2015 – June 2019 Ankara, Turkey

EXPERIENCE

Applied Scientist 2 <i>Microsoft</i> Confidence Estimation in LLMs: Developed post-training methods for calibrated confidence estimation in large language models, using auxiliary predictors and RL-based optimization. Our work resulted in improved hallucination detection and reliability in production settings such as Vision Language Models for structured extraction from long documents. LLM Post-training: Contributed to post-training of GPT-4o for multilingual settings across mid-training, SFT, DPO, and RLHF stages. Curated large-scale benchmarking datasets, implemented scalable evaluation pipelines, and developed synthetic data generation pipelines. Our work resulted in SOTA LLMs in multilingual settings. Azure AI Foundry Agents: Designed and optimized agentic workflows with tool calling, retrieval, and search, generating synthetic evaluation data and implementing benchmarking pipelines. Performed prompt engineering for the tool calling stage which achieved 30% faster responses without compromising AI quality.	May 2024 – Present Mountain View, CA
Postdoctoral Scholar Research Assistant <i>Stanford University</i> - Developed memory-efficient training techniques for high-dimensional MRI reconstruction with unrolled neural networks - Developed a self-supervised learning based augmentation framework for MRI reconstruction that improves robustness to distribution shifts such as signal-to-noise ratio change and motion corruption - Worked on equivalent convex formulations to non-convex neural networks to enable robustness, interpretability, and convergence guarantees	March 2024 – May 2024 Sep 2019 – Jan 2024 Stanford, CA
Research Intern <i>NVIDIA</i> - Worked on improving the robustness of diffusion models under distribution shifts for MRI reconstruction using Stein's Unbiased Risk Estimator (SURE) under the supervision of Dr. Chao Liu and Prof. Jan Kautz - Worked on improving the efficiency of diffusion models for MRI reconstruction using variational inference under the supervision of Dr. Morteza Mardani and Prof. Jan Kautz	Sep 2022 – April 2023 Santa Clara, CA
Research Intern <i>Microsoft Research</i> - Worked on identifying failure modes of in-context learning with Large Language Models (LLMs) - Developed a probabilistic inference technique for LLMs to improve their reasoning capabilities and robustness to prompt design under the supervision of Dr. Nebojsa Jojic	June 2022 – Aug 2022 Redmond, WA

Research Intern

Computer Vision Laboratory at ETH Zurich

July 2018 – August 2018

Zurich, Switzerland

- Compared beamforming and displacement tracking methods for imaging speed-of-sound with pulse-echo ultrasound under the supervision of Prof. Orcun Goksel

SELECTED PUBLICATIONS

1. Sahiner, A., Ergen, T., **Ozturkler, B.M.**, Pauly, J., Mardani, M. and Pilanci, M. Scaling Convex Neural Networks with Burer-Monteiro Factorization. **ICLR 2024**. <https://openreview.net/forum?id=ikmuHqugN7>
2. **Ozturkler, B.M.**, Mardani, M., Vahdat, A., Kautz, J., Pauly, J. RED-diff: Regularization by Denoising Diffusion Process for MRI Reconstruction, **NeurIPS 2023 Workshop on Deep Learning and Inverse Problems**
3. **Ozturkler, B.M.**, Liu, C., Eckart, B., Mardani, M. Song, J., Kautz, J. SMRD: SURE-based Robust MRI Reconstruction with Diffusion Models, **MICCAI 2023 (early accept, top 15% of all submissions)**
4. **Ozturkler, B.M.**, Malkin, N., Wang, Z. and Jojic, N. ThinkSum: Probabilistic reasoning over sets using large language models, **ACL 2023**. <https://arxiv.org/abs/2210.01293>
5. Sahiner, A.*, Ergen, T.*, **Ozturkler, B.M.**, Bartan, B., Pauly, J., Mardani, M. and Pilanci, M. (* equal contribution) Hidden Convexity of Wasserstein GANs: Interpretable Generative Models with Closed-Form Solutions. **ICLR 2022**. <https://arxiv.org/abs/2107.05680>
6. Ergen, T.*, Sahiner, A.*, **Ozturkler, B. M.**, Pauly, J., Mardani, M., Pilanci, M. (* equal contribution), Demystifying Batch Normalization in ReLU Networks: Equivalent Convex Optimization Models and Implicit Regularization, **ICLR 2022**. <https://arxiv.org/abs/2103.01499>
7. Sahiner, A., Ergen, T., **Ozturkler, B.M.**, Pauly, J., Mardani, M. and Pilanci, M. Unraveling Attention via Convex Duality: Analysis and Interpretations of Vision Transformers. **ICML 2022**.
8. Desai, A.D.*, Gunel, B.*, **Ozturkler, B.M.**, Beg, H., Vasanaawala, S., Hargreaves, B.A., Ré, C., Pauly, J.M. and Chaudhari, A.S. (* equal contribution) VORTEX: Physics-Driven Data Augmentations for Consistency Training for Robust Accelerated MRI Reconstruction. **MIDL 2022 Best paper award**. <https://arxiv.org/abs/2111.02549>, 2021
9. **Ozturkler, B.M.**, Sahiner, A., Ergen, T., Desai, A.D., Pauly, J.M., Vasanaawala, S., Mardani, M. and Pilanci, M. Greedy Learning for Large-Scale Neural MRI Reconstruction. **NeurIPS 2021 Workshop on Deep Learning and Inverse Problems, oral presentation**. <https://openreview.net/forum?id=tKeEIFvmENy>
10. Sahiner, A., Mardani, M., **Ozturkler, B. M.**, Pilanci, M., Pauly, J. Convex Regularization Behind Neural Reconstruction, **ICLR 2021**. <https://arxiv.org/abs/2012.05169>

TECHNICAL SKILLS

Languages: Python, MATLAB, R, C/C++

Deep Learning Libraries: PyTorch, TensorFlow, Keras, Jax

Other: Linux, Git, L^AT_EX, Weights & Biases, Docker, Google Cloud Platform, Microsoft Azure, Kubernetes

PROFESSIONAL ACTIVITIES

- **Reviewer:** ACL, Neurips, ICML, ICLR, MICCAI, IEEE Transactions on Medical Imaging,
- **Program Committee Member:** ICML Workshop on Knowledge and Logical Reasoning in the Era of Data-Driven Learning
- **Talks:** Greedy Learning for Large-Scale Neural MRI Reconstruction, NeurIPS 2021 Workshop on Deep Learning and Inverse Problems
- **Mentorship:** Mentored undergraduate students with the Caltech Summer Undergraduate Research Fellowships program

AWARDS AND HONOURS

- **National Science Foundation, Graduate Research Fellowship (NSF GRFP)** Three years of financial support and a cost of education allowance to the institution.
- **Stanford University Electrical Engineering Departmental Fellowship** Full tuition waiver & stipend during the first year of the Ph.D. program.
- **Ranked 197th nationally** in the University Entrance Examination in Turkey Among 2 million participants.